MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1DSCECT101 - COMPUTER FUNDAMENTALS AND BASICS OF PC HARDWARE

Duration: 1.5 hours Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	Identify the primary function of the Central Processing Unit (CPU) from the given list. a) Data Storage b) input/output operations c) Executing instructions d) Networking	U	1
2	Identify the primary function of the ENIAC computer. a) To perform simple calculations b) To solve complex mathematical problems c) To store data d) To run commercial applications	U	1
3	Identify the no: of unique characters that can be represented with 7 bits in ASCII. a) 64 b)128 c)256 d)512	U	1
4	Name the device which is typically used to connect different networks in a WAN. a) Hub b) Switch c) Router d) Repeater	K	1
5	Identify the act of retrieving data from a memory storage location a) Memory Write b) Backup c) Formatting d) Memory Read	U	2
6	Classify the secondary storage device best suited for long-term archiving of data. a) RAM b)SSD c)HDD d)Optical Disc	U	2
7	Indicate which option identifies the primary function of a mouse. a) To enter text b) To move the cursor and select items on the screen c) To print documents d) To process images	U	2
8	Compare the advantages of SSDs over HDDs in data retrieval during multitasking. a) HDDs retrieve data faster during multitasking due to their larger capacity. b) SSDs and HDDs offer similar performance in multitasking environments. c) SSDs have faster data retrieval, enhancing multitasking performance. d) HDDs have better performance due to their mechanical components.	An	3
9	Examine the effects of using multiple GPUs in a system and identify the primary advantage. a) It reduces the need for cooling solutions b) It enhances performance of graphics c)It simplifies the system architecture d) It decreases power consumption	An	3
10	Examine which type of cache writing strategy updates data in both the cache and the main memory simultaneously. Cache overflow b) Cache hit c) Cache miss d) Cache bypass	An	3

1 * 10 = 10

Part B

Short Answer Questions

Answer **4** Questions

Each question carries 5 marks

11	Discuss why memory management function of operating system is important.	U	1
12	Distinguish between hardware and software in a computer system.	U	1
13	Explain what registers are. Discuss the important registers used in CPU.	U	2
14	Describe the different types of printers available and identify the advantages and disadvantages of each type.	U	2
15	Explain Green Computing with an example.	An	3
16	Predict the potential benefits of neuromorphic computing in replicating human brain functions.	An	3

4 * 5 = 20

Part C

Essay Questions

Answer **2** Questions

Each question carries 10 marks

17	Describe the components of a computer system with a simple diagram to support your explanation.	U	1
18	Describe how the core functions of OS contribute to the overall system performance.	U	1
19	Explain the differences between various types of USB ports & CPU Sockets.	U	2
20	Differentiate Virtual Reality (VR) and Augmented Reality (AR) in terms of user experience and applications.	An	3

2 * 10 = 20